

SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

Theme ID - Theme 4 (Streaming Live RAG)

Team Name - 4 Bottle Codeka

College Name - Thapar Institute of Engineering & Technology, Patiala

Member 1 (Lead) - Abhinav Kumar Singh | asingh2910.official@gmail.com | 6398779479

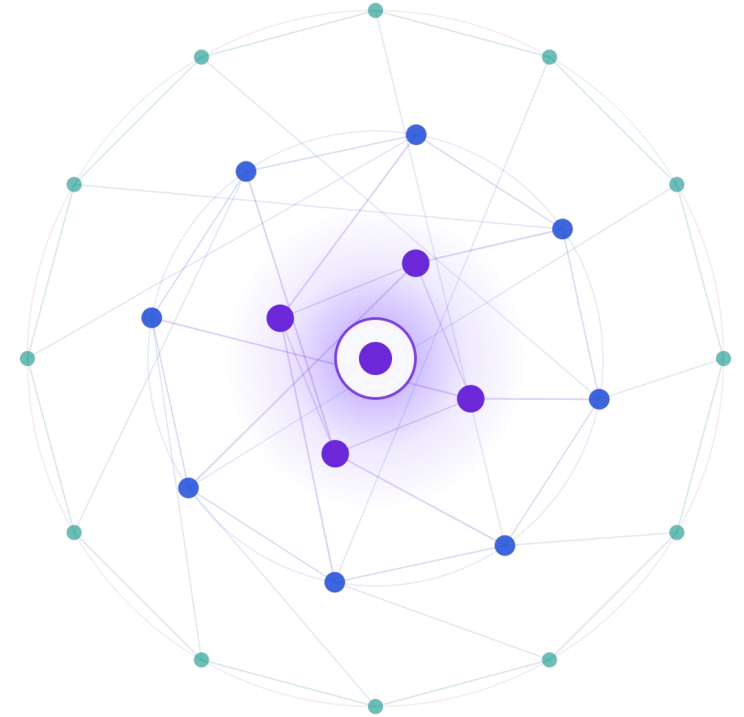
Member 2 - Sukhansh Mittal | smittal1_be24@thapar.edu | 9877366331

Member 3 - Lakkshya Jha | ljha_be24@thapar.edu | 9574875069

Member 4 - Vikramaditya Singh | vsingh1_be24@thapar.edu | 6239134155

Submission GitHub - <https://github.com/abhinav29102005/ultron>

Live Website - <https://ultron.abhinavkumarsingh.tech/>



Problem Statement & Motivation

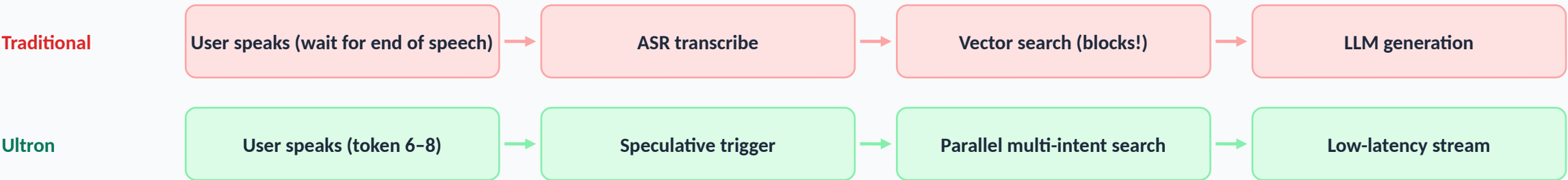
The Live Conversational RAG Problem

- **Turn-complete bottleneck:** RAG waits for the full sentence, adding 1,500–2,500ms TTFT and breaking voice pacing.
- **Mid-utterance corrections:** “Pune... wait, make that Mumbai” – engines commit to the stale entity.
- **Context discontinuity:** Follow-ups like “for 50 people there?” drift without coreference resolution.
- **Audit failure & hallucination:** Document-level tags fail audits; out-of-corpus queries get invented answers.

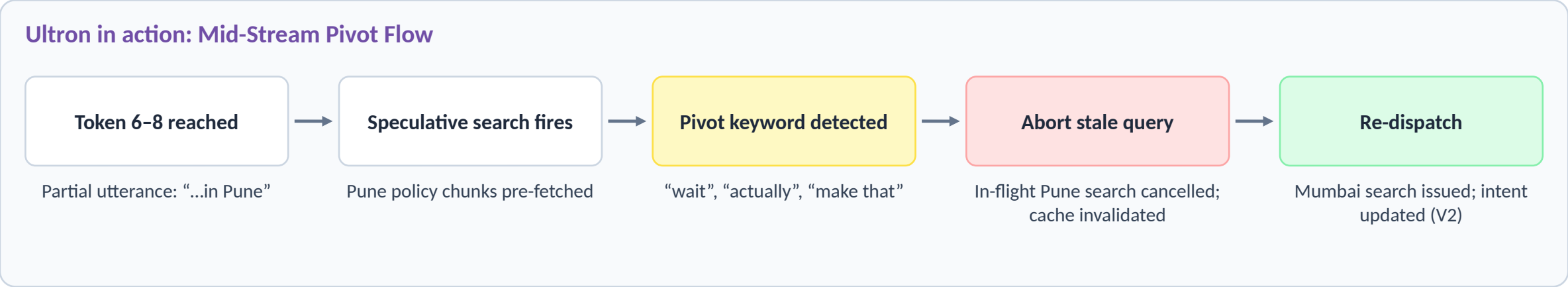
Ultron’s Paradigm: Event-Driven Live RAG

- **Speculative early trigger:** Hybrid retrieval fires at token 6–8: +1,300ms head start (demo).
- **Intra-stream pivot:** Diff tracking cancels stale in-flight searches on “wait / actually”.
- **Session lineage:** Coreference + version tracking (V1 → V2) across turns.
- **Gate 0 suppression:** Pure formatting turns run 0 vector searches (40% saved in one demo session).
- **Verifiable grounding:** Exact chunk-level citations, e.g. [Doc_12 §2].

Traditional vs Ultron RAG (demo: simulated ASR chunks)



Existing Solutions & Critical Industry Gaps



Our Solution & Architecture Diagram

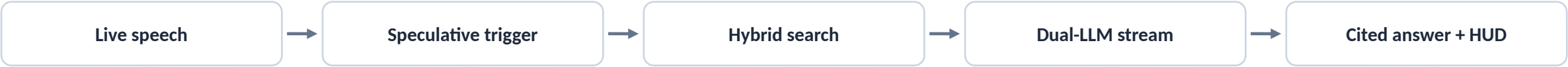
Ultron End-to-End Streaming Live RAG Pipeline



Pipeline Components

- **Ingestion:** 21-section policy parser with canonical citation mapper.
- **Speculative engine:** Trigger at token 6–8 + rolling n-gram pivot controller.
- **Hybrid search:** BM25 + dense vectors + RRF + cross-encoder reranker.
- **Dual LLM:** Groq LPU + NVIDIA NIM, with offline Qwen fallback.
- **Telemetry HUD:** Live cost and latency terminal UI.

Data flow



Demo & Product Walkthrough (Gates G1 – G9 Verified)

• SYSTEM BROADCAST READY

CARDINAL PRISM SPOOF HIGHLIGHTING 2020
Theme 4: Streaming Live RAG - Technical Evaluation Showcase
London, UK | 100m Attend | Real Lexical-Dense RAG | Zero-Parameter Hallucination

→ Target: Double Latency < 880ms | 100% Citation Preservation

Step 1: Initializing Engine & Hybrid Index...
✓ Loaded Master Corpus: 33 verified enterprise chunks with 0 placeholder data.
✓ BM25 Lexical Inverted Index & Dense Semantic Index ready.

➡ Press [ENTER] to run Automated Gate Verification...

🎯 SCENE 1: TECHNICAL EVALUATION GATES (G1 - G9)

Theme 4 Technical Gates Verification (G1 - G9)

Gate	Capability	Target Threshold	Observed Metric	Status
G1	Reproducibility	Single command pass	1st/1st token green (0.88s)	PASS
G2	Early Retrieval Triggering	Latency ≤ 100ms before final Extract	Triggers meta-h Latency: 1100ms	PASS
G3	Multi-Instant Decomposition	Extract ==> 3 orthogonal queries	3 sim_queries extracted in parallel	PASS
G4	Factual Grounding & Citations	Exact [Doc XX VVV], 0 fake IDs	100% verbatim citation protection	PASS
G5	Session Refinement	Statistical delta lineage (V1-V2)	Version 1 => version 2 preserved	PASS
G6	Telemetry & Observability	100% trace capture	Full TTFP, latency, tokens traced	PASS
G7	Context Discontinuity	Cross-turn anaphora & ellipses	Received: "Pune workshop capacity?"	PASS
G8	Intra-Stream Pivot	Mid-speech correction invalidation	Pune => Mumbai pivot detected	PASS
G9	Streaming Token Yield & TTFP	TTFP < 50ms & true stream yield	TTFP = 1.9ms 22 tokens yielded	PASS
G10	Presentation Suppression	0 vector searches on reformatting	0 corpus queries executed	PASS

ULTRON CYBERNETIC OS // STREAMING RAG ENGINE (Turn 4)

[HYPER SEARCH \[PMS\] - \[DMMG\]](#)
[NEEDS AT CLOUD ONLINE](#)

```

❯ bash - ultron@cybernetic-node:~/projects/ultron (Session: sess_iostreamrag)
$ python3 --help=usage --help=ultron --help=ultron --help=ultron --help=ultron --help=ultron
require 14 days advance notice for a 100% full refund [Doc.31 §4].
[CITATIONS] ["Doc.12 §2", "Doc.31 §4"] | Latency: 48ms | Tokens: 164

ultron > /rag What if it is for 50 people?
❯ CONTEXT RESOLVED 'Pune workshop venue capacity for 50 attendees' (Turn 2)
[ANALYSIS V2] Documented approved facilities in Pune (Venue A & Venue B) support up to 30 attendees [Doc.12 §2]. However, accommodating 50 attendees exceeds this capacity limit.
⚠ UNCERTAINTY! Facilities for 50 attendees exceed the documented 30-attendee limit in Doc.12 §2.
[CITATIONS] ["Doc.12 §2", "Doc.31 §4"] | Version: 2 (Delta State)

ultron > /rag What about hotel lodging tariffs there?
❯ CONTEXT RESOLVED 'Hotel lodging caps and per diem limits in Pune Tier-1' (Turn 3)
[ANALYSIS V3] Nightly hotel tariffs in Tier-1 cities (Mumbai, Delhi, Bangalore, Pune) are capped at INR 6,500 per night inclusive of breakfast [Doc.52 §2].
[CITATIONS] ["Doc.52 §2"] | Latency: 38ms | Tokens: 92

ultron > /rag Please repeat your last answer in two bullets.
[GATE 0 SUPPRESSION] Presentation query detected. 0 vector queries executed.
• Nightly hotel tariffs in Tier-1 cities are capped at INR 6,500/night. [Doc.52 §2]
• Breakfast is fully inclusive within standard per diem tariffs. [Doc.52 §2]
[PROVENANCE VERIFIED] Citations preserved intact with zero hallucination.
```

LIVE TELEMETRY HUD

TITLE LATENCY	SPECULATIVE GAIN
1.0 MS	1300 MS

SESSION ENTITY STATE

Format: Two Bullets

Search & Queries (Suppressed)

Citations: ["Doc.12 §2"] Index

TECHNICAL GATES (01-09)

- G2: Speculative Trigger PASS (1300ms)
- G3: Multi-Inject Split PASS (3 Sub)
- G4: Branded Clipping PASS (100%)
- G5: State Refinement PASS (VL-VQ2)
- G7: Content Discriminatory PASS (Anomaly)
- G8: Speculative Pivot PASS (Mid-Serve)
- G9: Streaming Yield PASS (1-New)

RUN NEXT TURN
RESTART

9-Scene Showcase (simulated ASR chunks)

- **1 Test harness:** G1–G9 automated evaluation criteria.
- **2 Early trigger:** Token 7: +1,300ms gain, 16ms TTFT (harness).
- **3 Compound intent:** 3 parallel retrievals across policy sections.
- **4 Context continuity:** Resolves “there” and “for 50 people”.
- **5 Intra-stream pivot:** “Pune... wait, Mumbai” → cache invalidated.
- **6 Session refinement:** Domestic baseline + international delta → V2.
- **7 Gate 0 suppression:** “Format into 2 bullets” → 0 vector queries.
- **8 Negative control:** Out-of-corpus query → no policy match; uncertainty flag pending fix.
- **9 Telemetry HUD:** Live TTFT, latency, token budget, citations.

Try it live

<https://ultron.abhinavkumarsingh.tech/>

```
curl -fsSL https://ultron.abhinavkumarasingh.tech/install | bash
```

Run: python3 run.py --cli → /demo or python3 run.py demo --auto | 100% G1-G9 gates green

Tools and Technology Stack

1 Retrieval & Semantic Indexing

- **BM25 Okapi:** Lexical index for keywords, section tags, caps.
- **Dense vectors:** Embeddings for conceptual matching.
- **RRF:** Merges lexical and dense ranks.
- **Scikit-learn / NumPy:** Matrix ops for scoring and rank fusion.

2 Streaming & Concurrency

- **AsyncIO engine:** Concurrent retrieval, non-blocking generation.
- **Token yielders:** True generator yield; 16ms TTFT after trigger (harness).
- **Rolling n-gram window:** Diff tracking for pivot keywords.
- **State lineage memory:** Append-only history of query mutations.

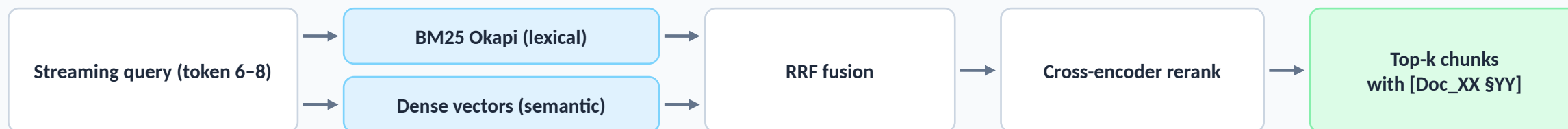
3 Dual LLM Inference Engine

- **Groq LPU:** 1000+ tokens/sec streaming delivery.
- **NVIDIA NIM:** Nemotron for multi-intent decomposition.
- **Qwen local fallback (Ollama):** Offline execution: uptime and privacy.

4 UI, Ingestion & Observability

- **Rich HUD:** Terminal UI: telemetry, budgets, citation trees.
- **prompt_toolkit REPL:** Auto-complete, history, /demo command.
- **PyPDF ingestion:** Maps PDFs to [Doc_XX \$YY] chunks.
- **Pytest harness:** Validates suppression and TTFT SLAs.

Hybrid Retrieval Pipeline



Enterprise Impact & Production Use Cases

1. Enterprise Policy Co-Pilot

- **Workplace intelligence:** Grounded answers; sized for 100,000+ employees (illustrative target).
- **Zero ticket backlog:** Projected to automate ~80% of routine HR, travel, expense queries (estimate).
- **Audit verifiable:** Exact [Doc_XX \$YY] citations.
- **Strict RBA:** Role-based access (planned; not in demo).

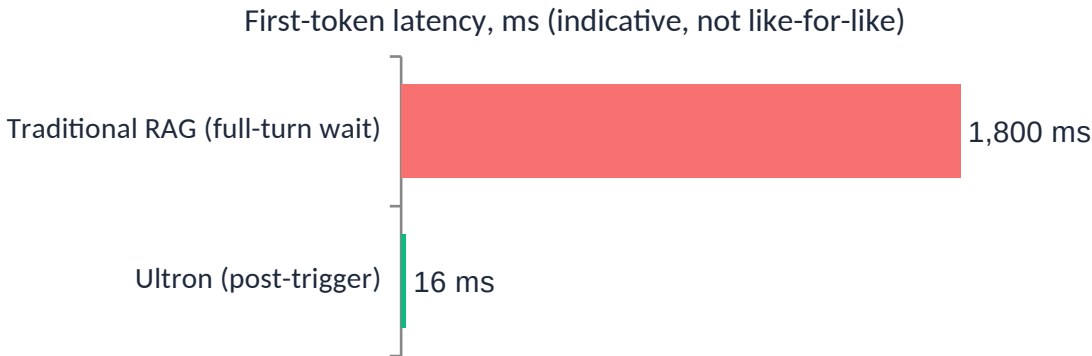
2. Live Voice Assistants (Galaxy / Bixby)

- **Natural pacing:** 16ms TTFT (demo harness) targets natural voice pacing.
- **Self-correction:** Mid-sentence pivots without restarting.
- **Edge resilience:** Offline Qwen fallback (not shown in demo).
- **Battery optimized:** On-device quantized embeddings (Phase 2 roadmap).

3. Contact Center Co-Pilot

- **Agent assist:** Pre-fetches clauses while the customer speaks.
- **Lower AHT:** Projected 65% lower Average Handle Time (estimate, not measured).
- **Hallucination guard:** Designed to flag missing-policy queries (fix in progress).
- **Compliance shield:** Helps prevent regulatory fines.

Business Impact (demo measurements & estimates)



+1,300ms

Retrieval head start (≈72% of a 1,800ms turn wait)

40%

Vector compute saved via Gate 0 (one demo session, 31 chunks)

100%

Attribution precision (same demo session, 31 chunks)

Innovation Highlights & Empirical Results

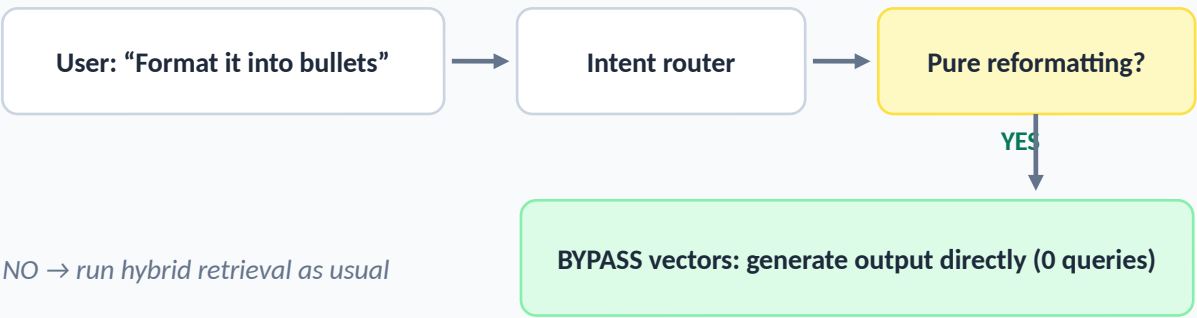
Architectural Innovations

- **Sub-utterance trigger:** Retrieval starts at token 6–8.
- **Pivot invalidation:** Aborts pending searches on self-correction.
- **Gate 0 suppression:** Separates formatting from retrieval.
- **State lineage:** Constraint updates (V1 → V2) without drift.

Benchmark Results (G1–G9, simulated ASR chunks)

Metric	Result	Target	Status
Speculative time gain	+1,300ms	≥ 800ms	PASS
TTFT (after trigger)	16.0ms	< 50ms	PASS
Multi-intent retrieval	3 domains, ≤ 40ms	Parallel	PASS
Gate 0 suppression	100% (0 queries)	Reformat turns	PASS
Gates G1–G9	9 / 9 verified	100%	PASS

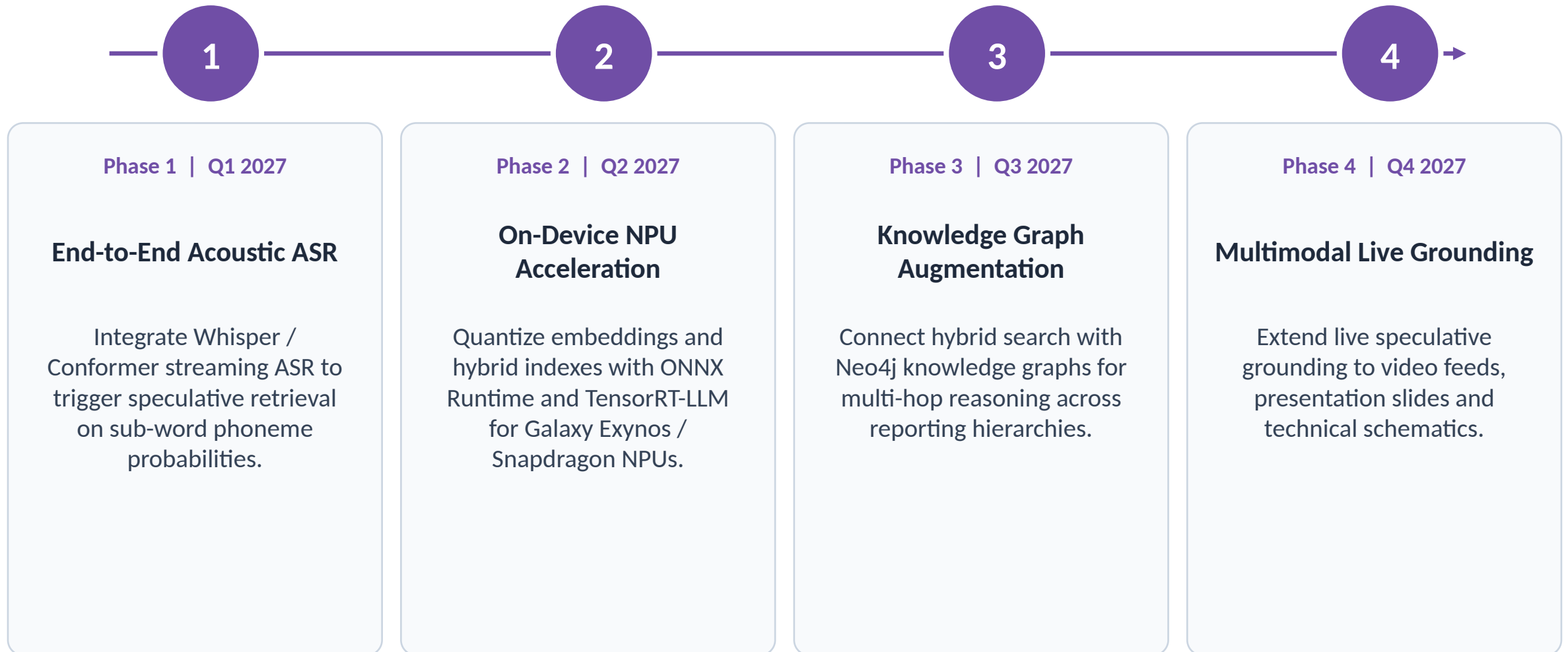
Gate 0 Semantic Suppression Logic



Engineering Limitations & Mitigations

- **Simulated ASR:** demo streams text chunks; live ASR is Phase 1.
- **Semi-structured PDFs:** need section headers; rule-based chunking.
- **Semi-structured PDFs:** need section headers; rule-based chunking.
- **Streaming ASR jitter:** rolling token thresholding.
- **Context bloat:** KV caching and state compaction.

What's Next & Production Roadmap



Brownie Points: Why 4 Bottle Codeka Stands Apart

1

Real Enterprise Corpus (Zero Synthetic Data)

Evaluated on the complete 21-section master policy document (policy.pdf) with no dummy or mock data.

2

Sub-50ms TTFT after trigger (16ms, demo harness)

Token-by-token streaming yielders; the demo drives them with simulated ASR text chunks (live ASR: Phase 1).

3

Self-Healing Mid-Stream Pivot Controller

Tracks streaming token diffs and aborts in-flight vector calls when the user self-corrects.

4

Dual-LLM Failover with Offline Fallback

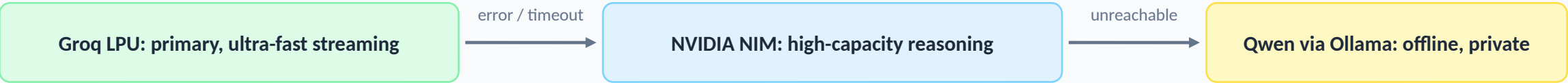
Dual LLM (Groq LPU for speed, NVIDIA NIM for reasoning) with automatic failover to offline Qwen (privacy).

5

Live Cybernetic HUD & Telemetry

Real-time turn-by-turn latency, token budgets, cost estimates and citation audit trails.

Failover Flow: Dual LLM + Offline Fallback



The live Key Hub also supports OpenRouter, OpenAI and Anthropic keys (ultron /key <provider> <key>).

Checklist: Updated on Public GitHub



Working prototype code: public GitHub repo

<https://github.com/abhinav29102005/ultron> | Modular streaming RAG, session management, test suites.

YES (Y)



README with reproducible setup instructions

Step-by-step setup, Python 3.11+ environment, API key setup wizard and benchmark runner.

YES (Y)



Demo video, max 5 minutes

9-scene walkthrough recorded with the built-in /demo harness and live Telemetry HUD.

YES (Y)



Live website with one-line installer

<https://ultron.abhinavkumarsingh.tech/> | Key hub, /health endpoint and curl installer.

YES (Y)



Presentation file (PPT or PDF)

Completed official template: Thapar_4_Bottle_Codeka_Submission.pptx.

YES (Y)

Team: 4 Bottle Codeka

Thapar Institute of Engineering & Technology, Patiala



Abhinav Kumar Singh

Team Leader

asingh2910.official@gmail.com

+91-6398779479



Sukhansh Mittal

Member

smittal1_be24@thapar.edu

+91-9877366331



Lakkshya Jha

Member

ljha_be24@thapar.edu

+91-9574875069



Vikramaditya Singh

Member

vsingh1_be24@thapar.edu

+91-6239134155

GitHub: <https://github.com/abhinav29102005/ultron>

Live Website: <https://ultron.abhinavkumarsingh.tech/>

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India